

KRISHNA CHARAN THOTA

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SUMMARY

Software Engineer with 4+ years building distributed systems and cloud-native services at Amazon and Armada AI using Go, Java, AWS, and Kubernetes. Delivered ingestion pipelines for performance metrics and fault alarms, secure webhook platforms, and observability systems across high-scale production environments. Experienced in owning services from design to deployment, debugging critical production issues, and driving reliability in fast-paced teams.

PROFESSIONAL EXPERIENCE

Software Engineer, Armada AI, Bellevue, WA

Mar 2025 – Present

- Designed and delivered a cloud-native data ingestion pipeline for **5G** and **satellite** infrastructure, integrating **Nokia** and **Starlink** APIs into a unified platform ingesting **performance metrics** and fault alarms at 15-second intervals.
- Built a secure **webhook** delivery system replacing high-volume email notifications with **JWT-authenticated** alert payloads to enterprise platforms like **ServiceNow**, including **whitelist validation** and end-to-end delivery.
- Resolved a critical silent data loss issue where **94 of 102 SIMs** and **non-RAN** alarms were silently dropped during **ingestion**, implementing a **fault-tolerant** fallback that achieved **100%** alarm ingestion in production.
- Optimized ingestion **throughput** through **asynchronous job orchestration** and **worker pool processing**, enabling reliable **15-second** metric ingestion cycles across distributed edge and IoT workflows.
- Developed **SIM** lifecycle management APIs supporting **100+** SIMs through full **v1→v2** migration and handling **vendor-specific** edge cases including partial success states and inconsistent SIM synchronization in production.
- Implemented **webhook** reliability using exponential backoff retry with a **15-minute** exhaustion window, **ScyllaDB**-backed delivery tracking, and automated **Mailgun** alerts to account admins, eliminating silent **webhook** failures.
- Engineered 5 dedicated **Prometheus** and **Grafana** dashboards across **5+** services, providing **full-stack** visibility into **system health**, **performance**, **reliability**, and **infrastructure metrics** covering **ingestion pipelines**, **vendor success rates**, **resource utilization**, and **real-time alerting** across distributed services.
- Integrated **Azure Key Vault** with Managed Identity for secret management and secure **token retrieval** workflows, eliminating hardcoded credentials and strengthening security compliance across services.

Software Development Engineer, Amazon Services LLC, Bellevue, WA

Oct 2022 – Mar 2025

- Architected **ETL** processes for **1.2 billion** metadata records, consolidating data into a centralized **Redshift** data warehouse, reducing reporting time by **27%**.
- Designed and optimized a high-performance **Java API** backed by **DynamoDB**, reducing data retrieval latency by **29%** and improving performance for production workloads.
- Built a **pre-production** environment with **AWS CodePipeline** and **CDK** in **TypeScript** to validate infrastructure changes before production, minimizing risks through continuous monitoring and logging.
- Executed a **throttling** configuration **limiting concurrent** requests during peak traffic, resulting in a **95%** reduction in service disruptions and preventing cascading failures under high load.
- Implemented automated **deployment rollback** with health check validation across **CI/CD** pipelines, enforcing **80%** code coverage threshold and reducing failed production deployments.
- Promoted code quality and maintainability through structured **unit testing** with **JUnit** and **Mockito**, maintaining **80%+** test coverage across critical service components.

SKILLS

Programming Languages: Java, Go, Python, TypeScript, JavaScript, Bash, HTML, CSS

Cloud & Databases: AWS (API Gateway, S3, SQS, CloudWatch, CloudFormation, IAM, ECS, Fargate, EC2, CDK), DynamoDB, Redshift, PostgreSQL, MySQL, ScyllaDB, Azure (Key Vault, Managed Identity, AKS, Workload Identity)

Frameworks & Libraries: Spring, Spring Boot, Spring Cloud, JUnit, Mockito, React.js, Redux, Jest, Cypress

Observability & Monitoring: Prometheus, Grafana, CloudWatch, OpenAPI/Swagger, Structured Logging

Security & Authentication: JWT (RS256), OAuth2, HMAC, Keycloak, Workload Identity Federation, Azure Key Vault

Containerization: Docker, Kubernetes, Helm, Istio

Development Tools: Git, GitHub, Maven, Jenkins, Jira, Asana, Agile/Scrum, CI/CD, QuickSight

APIs & Architecture: REST, Microservices, Distributed Systems, Event-driven architecture

CERTIFICATES

- [AWS Solutions Architect Associate](#)

EDUCATION

Masters of Business Administration, WestCliff University

Sep 2025 – Present

**Master of Professional Studies in Data Sciences and Applications,
University at Buffalo, State University of New York**

Aug 2021 – Aug 2022

**Bachelors, Computer Science and Engineering,
Vignan's Foundation for Science Technology and Research**

May 2016 – May 2020